

From OCR to Generative AI

Modernizing cargo and shipping document processing with Microsoft Azure AI

Executive Briefing

Volume 3 of 3

The Industry Challenge

Every parcel, pallet, and container is shadowed by 5 to 10 documents.

30–120

distinct document
types per major client

Thousands

of active vendor
layout templates

65–75%

straight-through
processing on legacy OCR

40–60%

of total cost spent on
exceptions and keying

The operational ceiling of legacy OCR is being tested.

Document variety is expanding, vendor layouts change without notice, mobile capture now dominates ingestion, and customers expect context-aware reasoning, anomaly detection, and natural-language summaries that GenAI has made the new baseline.

Where Legacy OCR Falls Short

Template brittleness

Every new layout breaks the extractor. Onboarding cycles run 3 to 8 weeks per layout family.

No comprehension

OCR returns characters, not meaning. It cannot validate that fields are mutually consistent.

Sensitive to capture quality

Mobile photography, wrinkles, glare, and rotation defeat engines built for flatbed scanners.

Handwriting and stamps lost

The most operationally important content — annotations, condition notes, signatures — is routed to manual keying.

Mixed-language fragility

Cross-border content arrives in mixed Latin, CJK, Cyrillic, and Arabic. Routing logic is itself brittle.

Stagnant data lake

Field extraction only. No semantics, no embeddings, no narrative — analytics value is bounded.

Compounded across hundreds of layouts and millions of monthly documents, these limitations consume 40–60% of total processing cost.

THE OPPORTUNITY

The Generative AI Shift

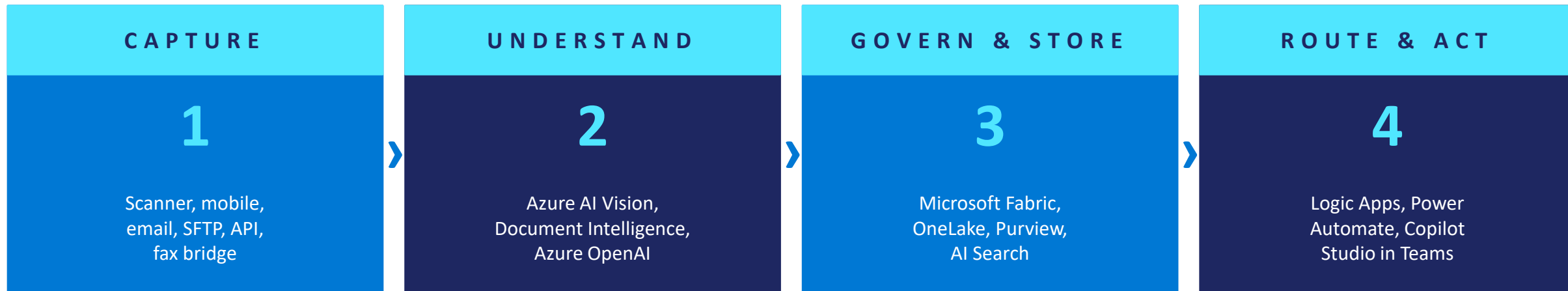
OCR extracts characters.

Generative AI extracts meaning.

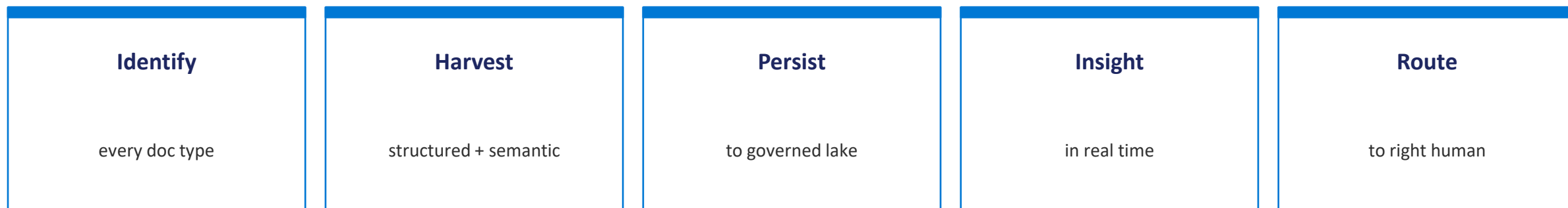
	Legacy OCR	Generative AI
Onboarding new layout	3–8 weeks	1–5 days
Straight-through processing	65–75%	90–95%
Cost per document (indexed)	100	50–70
Mobile capture acceptance	Limited	Native
Cross-document validation	Out of scope	Built-in
Multi-language documents	Routed pipelines	Single pass
Data lake depth	Field-level only	Field + narrative + graph

Targets are directional based on Microsoft published Document Intelligence and Azure OpenAI customer benchmarks.

Solution Vision — End-to-End Document Intelligence



CROSS-CUTTING — Microsoft Purview governance • Azure AI Content Safety • Active learning loop • Reasoning trace for audit



One pipeline. Every channel. Every document type. Every customer.

Microsoft Azure AI — Component Roles

1

Azure AI Vision

Image enhancement, baseline OCR, stamp and signature detection, image quality scoring.

2

Azure AI Document Intelligence

Custom and prebuilt extractors for AWBs, BOLs, invoices, weigh bills, customs forms.

3

Azure OpenAI Service

Multimodal reasoning, novel-layout extraction, cross-document validation, NL summaries.

4

Microsoft Fabric / OneLake

Unified data lake — bronze, silver, gold layers — with native Power BI Direct Lake.

5

Azure AI Search

Hybrid semantic index over the document lake — powers operator and customer copilots.

6

Logic Apps / Power Automate

Workflow routing, ERP/TMS integration, SLA timers, escalation, alerting.

7

Microsoft Copilot Studio + Teams

Operator copilot, customer SME copilot, customer self-service agent — all in Teams.

8

Microsoft Purview + Content Safety

PII classification, lineage, compliance, responsible-AI guardrails, audit.

Document Type Coverage

Document type	Primary Azure components	Key extracted entities
Air waybills (AWB)	AI Vision + Doc Intelligence custom	AWB#, shipper, consignee, weight, charges, route
Bills of lading	Doc Intelligence custom + Azure OpenAI	Container, vessel, ports, cargo description
Commercial invoices	Doc Intelligence prebuilt + Azure OpenAI	Line items, HS codes, COO, Incoterms, totals
Weigh bills	AI Vision + Doc Intelligence custom	Gross/tare/net, vehicle, route, scale ID
Customs declarations	Doc Intelligence custom + Azure OpenAI	Tariff codes, declared values, duties
Dangerous goods	Doc Intelligence custom + Azure OpenAI	UN#, packing group, hazard class
Proof of delivery (PoD)	AI Vision + Azure OpenAI	Timestamp, signatures, condition notes
Packing lists	Doc Intelligence custom	SKU, quantity, weight, dimensions

Coverage extends to certificates of origin, arrival notices, manifests, and vendor-specific variants — onboarded in days, not weeks.

Four Core Processing Capabilities

1 Document Identification

Classifier-driven routing replaces template matching. New layouts absorbed without breaking the pipeline. Bundled multi-doc captures auto-split.

2 Data Harvesting

Structured fields, line-item tables, and unstructured narratives extracted in one pass. Confidence scores per field drive downstream routing.

3 Data Lake Creation

Bronze, silver, gold medallion in Microsoft Fabric / OneLake. Power BI Direct Lake. Purview lineage and classification end-to-end.

4 Real-Time Insights

Anomaly detection, exception forecasting, compliance signals, and customer self-service intelligence — all generated in flight, not at month-end.

Workflow Routing and Next Best Action

Every document is routed by confidence, business rules, and cross-document checks — with an AI-generated next-best-action narrative attached.



EVERY ROUTED ITEM CARRIES AN AI-GENERATED NEXT-BEST-ACTION

"AWB 020-12345678 declares 4,500 kg — packing list sums to 3,200 kg. Recommended: hold for shipper confirmation. Auto-draft email below."

Conduent + Customer SME Engagement Model

CONDUENT OPERATIONS

Tier-1 Volume & Triage

- Assisted review of medium-confidence items
- Re-capture workflows for poor image quality
- Duplicate detection and resolution
- First-line customer communications
- SLA monitoring and operational reporting

SHARED WORK QUEUE

Microsoft Teams + Copilot Studio

- Unified queue surfaced to both sides
- Bidirectional escalation between tiers
- Source documents + validation history visible
- AI next-best-action on every item
- Every correction feeds weekly retraining

CUSTOMER SUBJECT MATTER EXPERTS

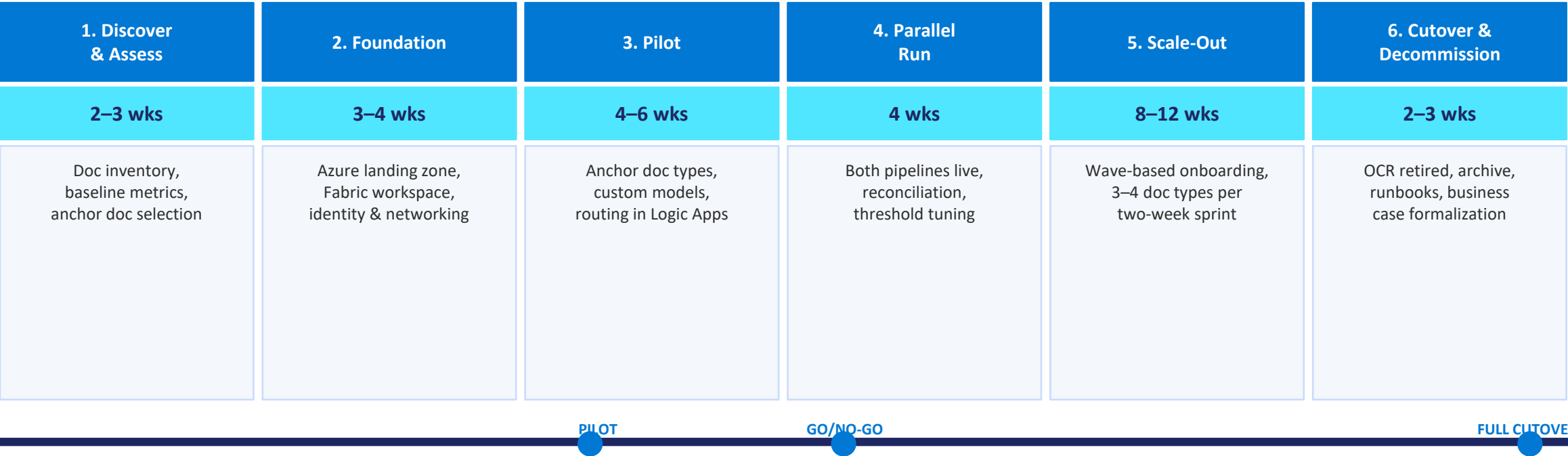
Tier-2 / 3 Adjudication

- Customs classification and valuation
- Dangerous goods reasoning and approval
- Business rule definition and evolution
- New shipper / lane onboarding
- Final approval on regulated submissions

Conduent runs the volume. Customer SMEs anchor the judgment. Microsoft AI amplifies both.

Conversion Plan — Legacy OCR to Azure AI

Phased, parallel-run delivery — production volume protected throughout. End-to-end: 6 to 8 months.



32 weeks total — vs 12 to 18 months for a traditional rebuild

Foundation models eliminate template engineering. Microsoft platform integration is pre-wired. AI-augmented engineering compresses every craft step.

Accelerated AI-Driven SDLC

The aggressive timeline is achievable because the entire delivery is itself executed on AI-augmented engineering tools.

GitHub Copilot

Application code — extractors, rules, integration adapters, BI calculations

Copilot in Azure

Infrastructure-as-code — Bicep / Terraform for landing zone and AI services

AI test generation

Unit and integration tests authored against real document samples

Synthetic data

Variant samples for regression where real data is scarce or sensitive

AI prompt engineering

Iterative prompt tuning for Azure OpenAI extractors against scored test sets

Story drafting

User stories and acceptance criteria drafted from document samples

Auto-doc

Runbooks, operator guides, customer-facing release notes

Active learning

Weekly retraining of custom Document Intelligence models from SME corrections

40–60% engineering velocity uplift on greenfield code

Microsoft published Copilot productivity benchmarks across enterprise development teams.

Business Value Realization

30–50%

**Cost reduction
per document**

Templating, exceptions, keying eliminated

Hours → Minutes

**SLA compression
on 90% straight-through**

End-to-end processing time

65% → 90%+

**STP rate uplift
on anchor doc types**

Validated in 4–6 week pilot

Weeks → Days

**New customer
onboarding cycle**

No more template authoring

Strategic

**Data lake becomes
an analytics asset**

Customer-facing intelligence

Continuous

**Active learning loop
from every correction**

System sharpens with use

Next Steps

1

Joint discovery workshop

Conduent and customer leadership validate document mix, anchor types, and target metrics.

2

Stand up Azure landing zone

Fabric workspace, identity, networking, and Document Intelligence in parallel — 3 to 4 weeks.

3

Run the 4 to 6 week pilot

Two or three anchor document types — typically AWBs, commercial invoices, and weigh bills.

4

Executive review at end-of-pilot

Confirm scale-out commitment based on validated accuracy and economics on live volume.

5

Phased scale-out and cutover

Wave-based onboarding of remaining document types — full conversion in 6 to 8 months.